

SOURAV YADAV

Blockchain Security Engineer | Smart Contract Security Researcher | Web3 Protocol Engineer

✉ sourav.dev.official@outlook.com | 💬 [WhatsApp](#) | 💬 [Telegram](#)

📍 Madhya Pradesh, India

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PROFESSIONAL SUMMARY

Smart Contract Security Researcher with hands-on experience auditing **21 production DeFi protocols and building blockchain security tooling**. Experienced in reasoning about complex protocol logic, protocol accounting, invariant analysis, exploit validation, and secure smart contract development.

Built protocol-aware security infrastructure and automated security tools in Solidity and Rust, with a strong interest in security research, complex code analysis, and advancing blockchain security.

TECHNICAL SKILLS

- **Smart Contracts:** Solidity, Foundry, Hardhat, OpenZeppelin, Proxy Patterns, Gas Optimization.
- **Security:** Smart Contract Auditing, Manual Code Review, Vulnerability Research, Fuzz Testing, Invariant Testing, Slither, Aderyn, Mythril, Tenderly, Exploit PoC Development.
- **DeFi:** AMMs, Uniswap V2/V3, Aave V3/V4, ERC4626 Vaults, Lending, Staking, Protocol Accounting, DEX Routing.
- **Web3 / Backend:** Ethers.js, Viem, Wagmi, Web3.js, The Graph, Chainlink, Node.js, Express.js, REST APIs, MySQL, PostgreSQL.
- **Frontend:** React.js, TypeScript, JavaScript (ESM), HTML5, CSS3, Tailwind CSS.
- **Developer Tools:** Git, GitHub, VS Code, Linux.
- **Languages:** Solidity, Rust, Python (Basic), C++, TypeScript, JavaScript, Java.
- **Networks:** Ethereum, Arbitrum, Base, Optimism, Polygon, BNB Chain, Solana.
- **Blockchain Explorers:** Etherscan, Arbiscan, Basescan, BscScan, PolygonScan.

SMART CONTRACT SECURITY EXPERIENCE

Independent Smart Contract Security Researcher

github.com/Sourav-IITBPL/audits

[Sherlock](#) | [Code4rena](#) | [Cantina](#)

July, 2025 – present

- Audited **21 production DeFi protocols**, covering AMMs, lending markets, ERC4626 vaults, staking protocols, and DEX infrastructure.
- Identified **11 validated vulnerabilities (2 High, 3 Medium, 6 Low)** with reproducible Proof-of-Concept exploit demonstrations.
- Specialized in **manual code review, protocol accounting analysis, invariant reasoning, attack surface analysis, exploit validation, and adversarial edge-case analysis**.

PROJECTS

PreFlight — Pre-Transaction Security Middleware

[GitHub Repository](#)

Solidity · Foundry · Chainlink CRE · React · Viem · Browser Extension · V2 Pools · ERC4626

- Built a **protocol-aware transaction security middleware** that analyzes DeFi transactions before execution using on-chain validation and off-chain simulation.
- Engineered hybrid risk validation combining **off-chain state simulation via Chainlink CRE** with **on-chain protocol guard contracts** for invariant and risk analysis.

- Implemented transaction decoding, NFT-based risk reporting, and execute/abort decision workflows; deployed across **Base, Ethereum, and Arbitrum testnets**.

Protocol Invariant Checker — DeFi Protocol Security Framework

[GitHub Repository](#)

Rust · CLI · Security Tooling · Protocol Analysis

- Built a **modular Rust framework** for **automated protocol invariant analysis, vulnerability detection, and structured smart contract security** reporting.
- Designed a protocol-agnostic architecture using a shared Protocol interface, enabling seamless integration of new protocol implementations.
- Generated standardized security reports with **severity classification, vulnerability categorization**, and **remediation recommendations**.

SettleOne — Yield-Generating Payment Commitment Protocol

[GitHub Repository](#)

Solidity · Foundry · Chainlink · React · NodeJs · ExpressJs · Viem · DeFi Yield Strategies · StableCoins

- Built a full-stack **payment commitment protocol** addressing delayed MSME/B2B payments through escrow-backed on-chain deal enforcement.
- Designed Solidity contracts for **deal lifecycle management, dispute arbitration, delivery verification, settlement, and fund release**.
- Integrated **yield-generating stablecoin escrow strategies** to improve capital efficiency during locked commitment periods; deployed across multiple testnets.

DexGateway — Multi-Chain DEX Routing Infrastructure

[GitHub Repository](#)

Solidity · Foundry · React · NodeJs · ExpressJs · TypeScript · The Graph

- Built full-stack DEX routing infrastructure integrating **7 Uniswap V2-style DEXes across various blockchain networks**.
- Implemented unified routing logic for **token swaps, liquidity provisioning, and liquidity withdrawal** through router abstraction.
- Engineered liquidity discovery, execution routing, and frontend transaction workflows across fragmented liquidity environments.

EDUCATION

Indian Institute of Information Technology (IIIT), Bhopal

Class of 2027

B.Tech — Electronics & Communication Engineering

CGPA: 7.57 / 10 | Expected Graduation: 2027

Maharishi Vidya Mandir, Mandla

Senior Secondary (Class XII) — **89.4%**

ACHIEVEMENTS

- **Cyfrin Updraft Certified** — Advanced Solidity development, smart contract security, protocol testing, and DeFi protocol engineering ([Certification Profile](#)).
- **LeetCode Knight (1868 Rating)** — Solved **500+ DSA problems** with strong algorithmic problem-solving skills.
- **HackVision 2026** — **Top 100 finalist team** among **400+ teams** in a national virtual hackathon.
- **Smart India Hackathon 2024** — National-level participant and team lead.
- **AlgoUniversity Tech Fellowship** — Selected for **2024 & 2025 cohorts**.
- **HER DAO Rust Cohort 1 Graduate**.